

Project Report Format

1.INTRODUCTION

1.1 Project Overview

The **Library Book Request Management System** is a ServiceNow-based application developed to automate the process of requesting library books. Students can submit book requests through an online form, while the system validates the issue date, generates a unique request number, stores the request in a custom table, and automatically notifies the librarian through email. The application replaces the traditional paper-based process with a secure and efficient digital workflow.

1.2 Purpose

- Automate library book request processing.
- Reduce manual paperwork.
- Validate issue dates automatically.
- Generate unique request numbers.
- Notify librarians through email.
- Maintain centralized request records.

2.IDEATION PHASE

2.1 Problem Statement

The existing library book request process is manual, making it time-consuming and difficult to track requests. There is no automatic validation of issue dates or notification to the librarian. The proposed system automates the complete request process using the ServiceNow platform

2.2 Empathy Map

Students need a simple and fast way to request books without paperwork. Librarians require an organized system that automatically validates requests, stores records, and sends notifications. The solution improves user experience by reducing delays and manual work.

2.3 Brainstorming

Several ideas were discussed before selecting the final project:

- Library Book Request Management System
- Employee ID Card Request
- Laptop Request Process
- Leave Approval System
- Visitor Management System
- Licensed Software Installation Request
- Automated Network Request Management
- Metro Ticket Generation System

After evaluating feasibility and usefulness, the **Library Book Request Management System** was selected.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The student accesses the request form, enters book details, submits the request, and receives confirmation. The librarian is automatically notified and processes the request.

3.2 Solution Requirement

The system should provide a digital request form, validate issue dates, generate unique request numbers, store records securely, and notify the librarian through email.

3.3 Data Flow Diagram

The student submits a request through the ServiceNow form. The Business Rule validates the issue date, generates a request number, stores the record in the custom table, and triggers Flow Designer to send an email notification to the librarian.

3.4 Technology Stack

- Platform: ServiceNow Zurich
- Database: ServiceNow Custom Table
- Validation: Business Rule
- Workflow: Flow Designer
- Notification: ServiceNow Email Notification
- Programming Language: JavaScript

4. PROJECT DESIGN

4.1 Problem–Solution Fit

The proposed solution replaces the manual library request process with an automated ServiceNow application that validates requests, generates request numbers, stores records, and sends notifications.

4.2 Proposed Solution

A custom ServiceNow application was developed with a request form, Business Rule, Flow Designer, and email notification to automate the entire book request workflow.

4.3 Solution Architecture

The system consists of a ServiceNow request form, Business Rule for validation, custom table for data storage, Flow Designer for workflow automation, and email notifications to the librarian.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was completed using Agile planning with two sprints.

- Sprint 1: Requirement Analysis and System Design
- Sprint 2: Development, Testing, and Documentation

The total project velocity was **16 story points per sprint**.

6. FUNCTIONAL & PERFORMANCE TESTING

6.1 Performance Testing

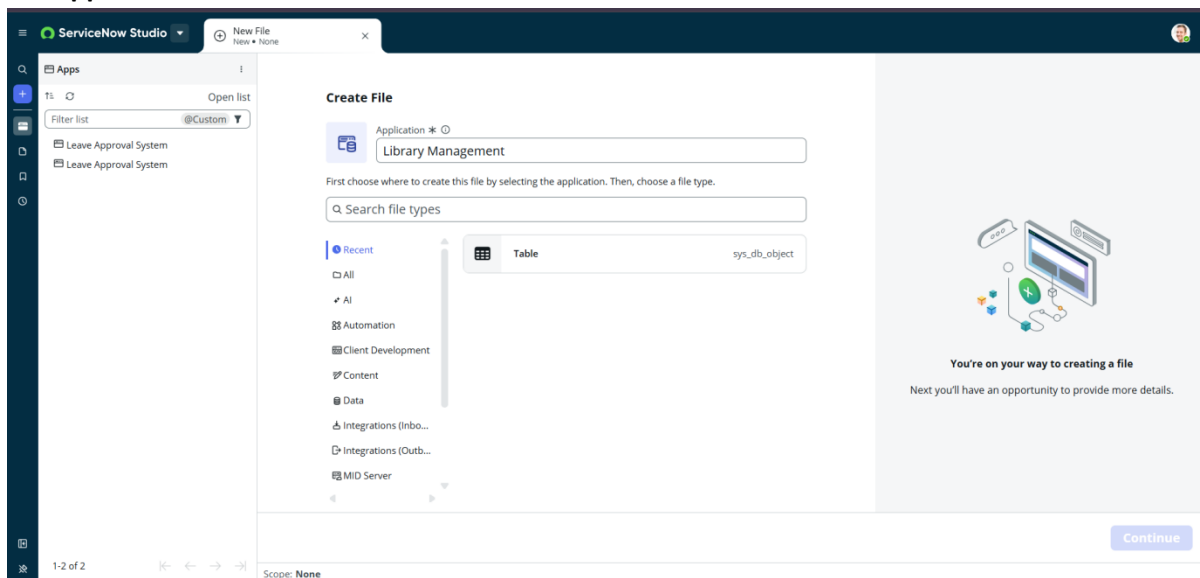
The application was tested for:

- Successful book request submission
- Issue date validation
- Request number generation
- Record storage
- Email notification

All test cases passed successfully.

7. RESULTS

7.1 Application Home



7.2 Book Request Form

The screenshot shows the 'Library Book Request - Create LIB10005' form in ServiceNow. The form includes the following fields:

- Number: LIB10005
- Student Name: Appalaraidu
- Student ID: 22IT001
- Book Name: Python Programming
- Author: Guido van Rossum
- Issue Date: 2026-06-26
- Status: Requested

A 'Submit' button is located at the bottom left of the form.

7.3 Business Rule

The screenshot shows the 'Business Rule' configuration page for 'Validate Issue Date'. The rule is set to run on 'Insert or Update' and is active. The 'When to run' tab is selected, showing the following configuration:

- When: before
- Order: 100
- Filter Conditions: (empty)
- Role conditions: (empty)

The 'Actions' tab is also visible, showing the following actions:

- Insert: ☒
- Update: ☒
- Delete: ☐
- Query: ☐

7.4 Past Date Validation

The screenshot shows the 'Library Book Request - Create LIB10009' form in ServiceNow. The form includes the following fields:

- Number: LIB10009
- Student Name: Somu
- Student ID: 22A13R4
- Book Name: IOT Systems
- Author: Faucon
- Issue Date: 2026-06-26
- Status: Requested

Two validation errors are displayed at the top of the form:

- Issue Date cannot be in the past.
- Invalid insert

A 'Submit' button is located at the bottom left of the form.

7.5 Submitted Request Record

The screenshot shows the ServiceNow interface with the 'Emails View: Outbox' selected. The table displays the following data:

Created	Recipients	Subject	State
2026-06-27 01:00:12	alileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	Ready
2026-06-27 01:00:09	alileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	Ready
2026-06-27 01:00:11	alileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	Ready
2026-06-27 05:06:40	appalanaiduchinni942@gmail.com	New Library Book Request Submitted	Ready
2026-06-27 00:00:11	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	Ready

7.6 Custom Table/List View

The screenshot shows the ServiceNow Studio interface for a 'Library Book Request' form. The form fields are as follows:

- Number: LIB10004
- Student Name: [Empty]
- Student ID: [Empty]
- Book Name: [Empty]
- Author: [Empty]
- Issue Date: YYYY-MM-DD
- Status: -- None --

7.7 Email Notification

The screenshot shows the 'Email Notification' configuration page for the event 'New Library Book Request Submitted'. The configuration details are as follows:

- Mailbox: Outbox
- State: Ready
- Target: Library Book Request: LIB10011
- User: [Empty]
- UID: N/A
- Created: 2026-06-27 05:06:40
- Deleted: [Unchecked]
- Weight: 0
- Importance: [Empty]

The 'Originating Event and Notification' section shows:

- Event: notification_engine.process
- Notification: Notify Librarian

The 'Subject' field is 'New Library Book Request Submitted' and the 'Recipients' field is 'appalanaiduchinni942@gmail.com'.

The 'Body' field contains the following HTML:

```
<html><head><style>
  #sync_notification_preference {
    text-decoration: none;
  }

  #sync_notification_unsubscribe {
    text-decoration: none;
  }
</style>
</head><body><p>A new library book request has been submitted.</p>
```

8. ADVANTAGES & DISADVANTAGES

Advantages

- Paperless process
- Faster request handling
- Automatic date validation
- Unique request number generation
- Email notification
- Secure record management

Disadvantages

- Requires internet connectivity
- Depends on the ServiceNow platform
- Limited to library request management

9. CONCLUSION

The **Library Book Request Management System** successfully automates the book request process using the ServiceNow platform. It validates issue dates, generates unique request numbers, stores request records securely, and automatically notifies the librarian. The system improves efficiency, accuracy, and user experience compared to the traditional manual process.

10. FUTURE SCOPE

- Book availability tracking
- Approval workflow
- SMS notifications
- Student self-service dashboard
- Fine calculation
- QR code integration
- Mobile application support

11. APPENDIX

Source Code

ServiceNow Application (Custom Table, Business Rule, Flow Designer, Notification)

Dataset Link

Not Applicable (No external dataset used)

GitHub & Project Demo Link

- GitHub Repository: <https://github.com/Appalanaidu12E1>
- Project Demo Video: https://drive.google.com/file/d/13PJF3_Zz1Fhde7pRmlQPee3vg-LX6f7j/view?usp=drive_link